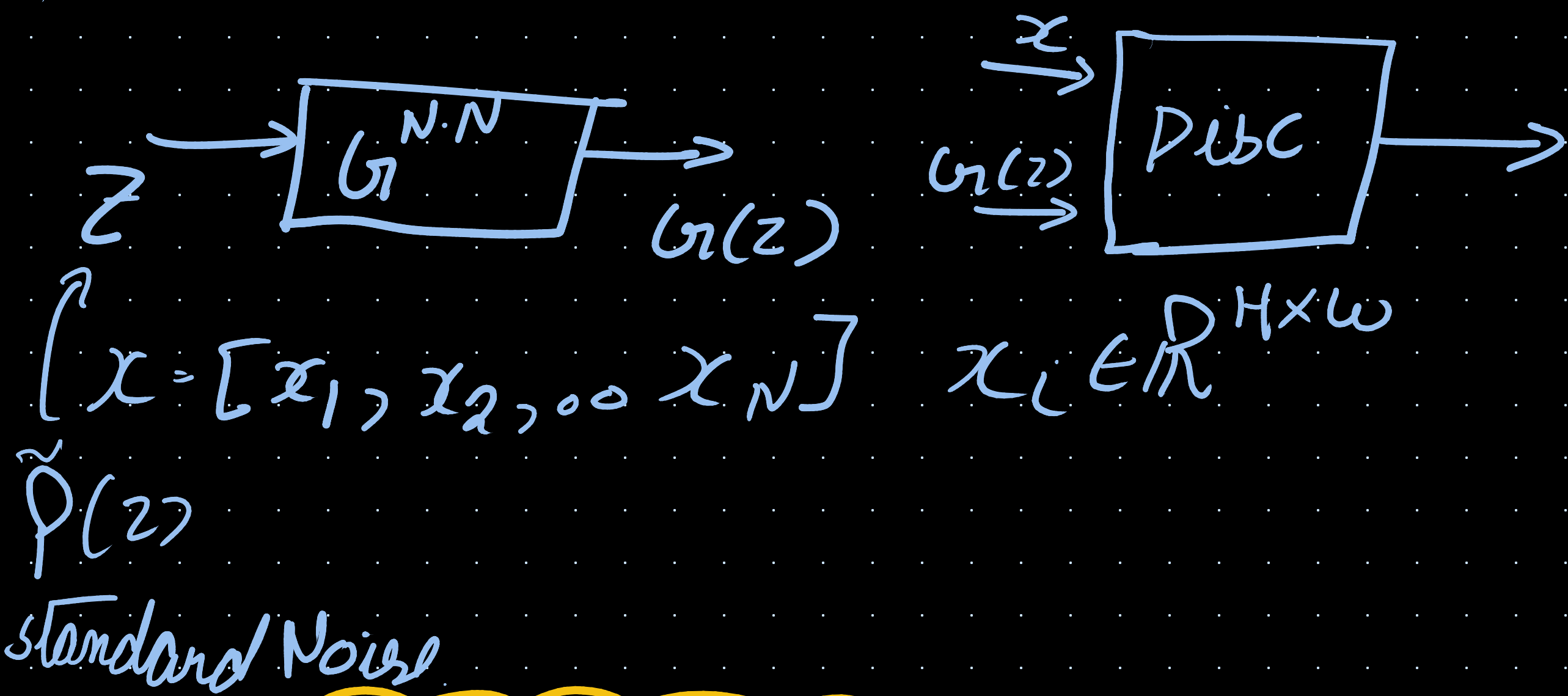


GAN



Objective
 $G(z)$ should follow $P_{True}(x)$

Discriminator's goal is to score the sample generated by generator.
 $\rightarrow$ Classifier to discriminate $P_{True}(x)$ & $G(z)$

Generator's goal is to fool the Discriminator.

Discriminator $\rightarrow$

$$-p \log p - (1-p) \log(1-p)$$

$$x \rightarrow [D_\theta] \rightarrow 1 \text{ (Real)}$$

$$G(z) \rightarrow [D_\theta] \rightarrow 0 \text{ (Fake)}$$

$$x \rightarrow \boxed{D_\theta} \rightarrow D(x)$$

Predicted Label.

$$G(z) \rightarrow \boxed{D_\theta} \rightarrow D(G(z))$$

$$\begin{aligned} BCE(D(x)) &= -1 \log D(x) - (1-1) \log(1-D(x)) \\ &= -\log D(x) \quad \text{--- (1)} \end{aligned}$$

$$BCE(D(G(z))) = -\log(1-D(G(z))) \quad \text{--- (2)}$$

$$\text{Loss of } D_\theta = \text{(1)} + \text{(2)} = -\log D(x) - \log(1-D(G(z)))$$

$$\min_{D_\theta} (-\log D(x) - \log(1-D(G(z))))$$

$$\max_{D_\theta} (\log D(x) + \log(1-D(G(z))))$$

Generator $\rightarrow$

$D(G(z))$ should be 1 (fooling D)

$\hat{p}$

$\hat{p}$

$$\text{BCE}(D(G(z))) = -\log[D(G(z))]$$

$$\min_{G, \phi} [-\log D(G(z))] \Rightarrow \max_{G, \phi} [\log D(G(z))]$$

$$\Rightarrow \min_{G, \phi} [\log(1 - D(G(z)))]$$

Loss $\rightarrow$

$$\min_{\phi} \max_{\theta} [\log(D(x)) + \log(1 - D(G(x)))]$$

MIN-MAX

Generator is not shown the P_{True} & is guided only with help of Discriminator.

If D_{θ} is perfect the G, ϕ would not converge.

So both should learn/work together.

In loss fun^c $\rightarrow$

$$\min_{\phi} [\log(1 - D(G(z)))] \Rightarrow \max_{\phi} [\log(D(G(z)))]$$

Does not saturate

gives better grad.

than $\min_{\phi} [\log (1 + G(z))]$

$$\min_{\phi} [\log D(G(z))]$$

→ This saturates at initial iterations.

D can easily discriminate u/w . $P_{\text{True}}(x) \neq G(z)$.

Here we use Jensen Shannon Divergence rather than KL Div.

Conditional

$(y \rightarrow x)$

$$\begin{bmatrix} \{x\} & \{y\} \\ D_1 & D_2 \end{bmatrix}$$

① class driven generation

② edge-Pixel

③ semantics

Model to map D_1 to D_2 & D_2 to D_1

Comparison u/w KL div & Jensen-Shannon Divergence

Divergence:

$$KL = \int P \log \left(\frac{P}{Q} \right) dx$$

More smoother than

KL D ??

$$JSD(P \parallel Q) = \frac{1}{2} D_{KL} \left(P \parallel \frac{P+Q}{2} \right) + \frac{1}{2} D_{KL} \left(Q \parallel \frac{P+Q}{2} \right)$$

$$[P(x) = 0; Q(x) > 0] \text{ \& \& } [P(x) > 0 \& Q(x) = 0]$$

→ JSD is const. for these cases

$$\max_{\theta_d} \min_{\theta_g} \left[E_{x \sim p(x)} [\log(D(x))] + E_{z \sim p(z)} [\log(1-D(G(z)))] \right]$$

$V(\theta_d, \theta_g)$

$$\rightarrow \int_x p(x) \log(D(x)) dx + \int_z \log(z) \log(1-D(G(z))) dz$$

$$\text{Optimal } D^* \rightarrow \max_{\theta_d} V(\theta_d, \theta_g)$$

Here we take x rather than z because it is in same space as true data

$$\rightarrow \int_x p(x) \log(D(x)) dx + \int_x \underbrace{p_g(x)}_{\text{estimated Density function of } G(z)} \log(1-D(x)) dx$$

$$\int_x [P(x) \log(D(x)) + P_g(x) \log(1-D(x))] dx.$$

D^* loss fun^c w.r.t θ_d .

$$f(y) = a \log y + b \log(1-y)$$

$$\frac{\partial f(y)}{\partial y} = \frac{a}{y} - \frac{b}{(1-y)} = 0$$

$$a - ay^* = by^*$$

$$y^* = \frac{a}{(a+b)}$$

$$D^* = \frac{P(x)}{P(x) + P_g(x)}$$

$$D^* = \frac{P_{\text{data}}(x)}{P_{\text{data}}(x) + P_g(x)}$$

Ideally $P_g(x) \sim P_{\text{data}}(x)$

$$D^* = \frac{1}{2}$$

Putting D^* in $V(\theta_d, \theta_g)$:

$$\int P_{\text{data}}(x) \log \frac{P_{\text{data}}(x)}{P_{\text{data}}(x) + P_g(x)} dx$$

↳ Training of G is complete & it is unable to discriminate b/w the generated & true

$$+ \int_x P_g(x) \log \left[1 - \frac{P_{\text{data}}(x)}{P_{\text{data}} + P_d(x)} \right] dx$$

$$\int_x P_{\text{data}}(x) \log \left(\frac{P_{\text{data}}(x)}{P_{\text{data}}(x) + P_g(x)} \right) + P_g(x) \log \left(\frac{P_g(x)}{P_d(x) + P_g(x)} \right)$$

Adding & sub $\log 4$

$$-\log 4 + \int_x P_d(x) \log \left(\frac{P_d(x)}{\frac{P_d(x) + P_g(x)}{2}} \right) + P_g(x) \log \left(\frac{P_g(x)}{\frac{P_d(x) + P_g(x)}{2}} \right) dx$$

$\rightarrow D_{KL}(P_d || \frac{P_d + P_g}{2})$

$\rightarrow D_{KL}(P_g || \frac{P_d + P_g}{2})$

$$= -\log 4 + 2 JSD(P_g || P_d)$$

$$C(b) = -\log 4 + 2 JSD(P_g || P_d)$$

$\rightarrow$ we train GAN using classifier, however

this shows that the process is equivalent to optimizing JSD!!

Earlier we said that if D is optimal D^* the model does not learn. However above derivation says otherwise (IN THEORY)

$$D_{KL}\left(P \parallel \frac{P+Q}{2}\right) + D_{KL}\left(Q \parallel \frac{P+Q}{2}\right) \\ \int P \log\left(\frac{2P}{P+Q}\right) + Q \log\left(\frac{2Q}{P+Q}\right) dx.$$

In case of non-overlapping distributions
 $JSD(P \parallel Q) = \log 2$

$$C(G) = -\log 4 + 2 \log 2 = 0$$

Towards principled methods for training GANs

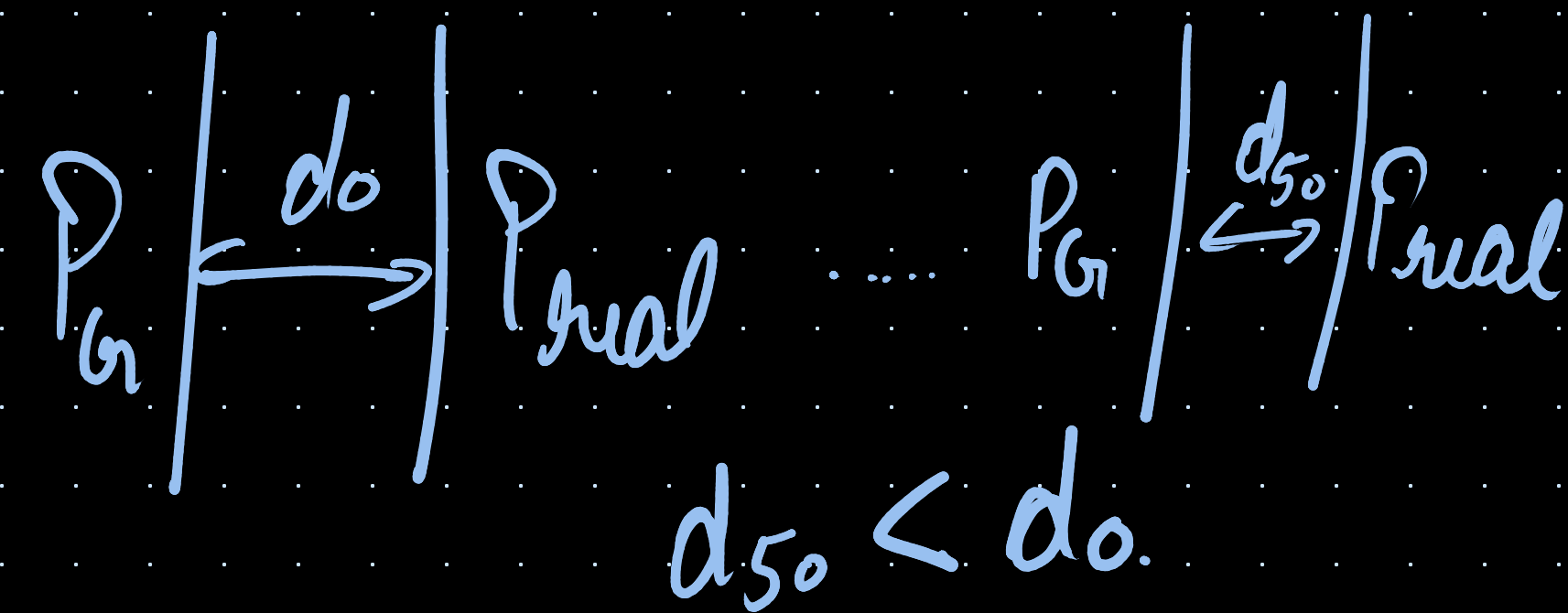
DCGAN paper $\Rightarrow$ JSD version is not known.

WGAN

Use Wasserstein / earth mover distance.
They removed sigmoid & use R value out of D directly.

We faced issue where non-overlapping distributions can result in constant loss making it impossible for model to learn.

To solve it we use earth mover / Wasserstein distance.



When a Discriminator learns to separate real and fake data effortlessly, its decision boundary becomes incredibly steep. A tiny change in the input image x results in a massive, unpredictable swing in the Discriminator's output $D(x)$. Mathematically, its gradients explode, and the Generator receives garbage signals.

To fix this, we need to enforce a "speed limit" on how fast the Discriminator's output can change with respect to its input. This concept is called Lipschitz Continuity.

Wasserstein distance is min. number of moves to reach Q distribution from P .

$$\text{Wasserstein} = W(P_A, P_B) = \inf_{\gamma \in \Gamma(P_A, P_B)} E_{(x,y) \sim \gamma} [\|x - y\|_2]$$

$$E[D(x)] - E[D(G(z))]$$

$$W(p_h, p_\theta) =$$

$$\max_{w \in W} \int_{x \sim P_h(x)} \int_{z \sim P_\theta(z)} \dots$$

Discriminator / critic network

For $x \rightarrow \max_w D_w(x) \uparrow \rightarrow$ we want it to tend to ∞

$\max_w -D_w(G(z)) \uparrow$

$\min_w D_w(G(z)) \downarrow \rightarrow$ we want to tend to $-\infty$

Discriminator learns to separate real & fake data.

$$\max_G [E(D(G(z)))]$$

Generator wishes to fool D.

$$\min_G [-E[D(G(z))]]$$

$$\text{loss} = \min_G \max_w \{ E_x [D_w(x)] - E_z [D_w(G(z))] \}$$

Dual form

Primal form = $W(P_r, P_g)$

WGAN works iff Discriminator is a Lipschitz function

il. $f(x) - f(y) \leq 1 - \|x - y\|_2$
Forcing Discriminator to be stable.

In original work the weights were clipped to enforce lipchitz.

WGAN-GP

$$L = \max_{\theta \in \Theta} \min_{w \in W} E_{z \sim p_z} [D_w(g_\theta(z))]$$

$$- E_{x \sim p_z} [D_w(x)] + \lambda E_{\hat{x}} (\|\nabla D_w(\hat{x})\|_2 - 1)^2$$

Gradient Penalty

Also helps keep loss from oscillating.

Spectral Normalization of GAN

Prevents D from becoming too powerful & helps in stabilizing.

At each layer of D_c

$$\hat{w} = \frac{w}{\sigma(w)}$$

$\omega \rightarrow$ parameters $\sigma(\omega) \rightarrow$ Spectral norm of ω

Look at all normalizations & check other usage of spectral ["].

Progressively growing GAN to be done

Q DeconvLayer, strided Layer, fractional strided Layer.